

Team name: RRR Kabelmüsli

Exchange Rate € → \$ 1,15

Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Euro EUR	1.106,00 €	\$1.271,90

#	Component	Part name	Autor	Source	Quantity	Unit cost (Local Currency)	Unit cost (US Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)
1	PCB with electronic components	Custom made	Simon Mütter	aisler.net	3	32,16 €	\$36,98	96,48 €	\$110,95
2	Motor	125:1 Metal Gearmotor 20Dx55L 12V CB with Extended Motor Shaft	Pololu	https://eckstein.com	4	43,97 €	\$50,57	175,88 €	\$202,26
3	Encoder	Pololu Magnetic Encoder Pair Kit for 20D mm Metal Gearmotors, 20 CPR, 2.7-18V 3499	Pololu	https://eckstein.com	2	11,84 €	\$13,62	23,68 €	\$27,23
4	Gyro	WT901	WITMOTION	https://www.witmotion.com	1	42,00 €	\$48,30	42,00 €	\$48,30
5	Microcontroller	Teensy 4.1	PJRC	https://www.pjrc.com	1	39,50 €	\$45,43	39,50 €	\$45,43
6	Single-board-computer	Raspberry Pi 5	Raspberry Pi	https://www.raspberrypi.com	1	135,95 €	\$156,34	135,95 €	\$156,34
7	5V Voltage Regulator	Pololu D24V50F5	Pololu	https://eckstein.com	1	39,21 €	\$45,09	39,21 €	\$45,09
8	5V Voltage Regulator	TSR 3-2450N	Traco Power	https://www.traco.com	1	7,26 €	\$8,35	7,26 €	\$8,35
9	3.3V Voltage Regulator	TSR 3-2433N	Traco Power	https://www.traco.com	1	7,95 €	\$9,14	7,95 €	\$9,14
10	12V Voltage Regulator	Pololu S13V25F12	Pololu	https://eckstein.com	1	22,55 €	\$25,93	22,55 €	\$25,93
11	Time-of-Flight distance sensor	VL53L1X	Pololu	https://eckstein.com	8	27,21 €	\$31,29	217,68 €	\$250,33
12	Multi-Target-Time-of-Flight sensor	VL5318CX	Pololu	https://eckstein.com	3	35,64 €	\$40,99	106,92 €	\$122,96
13	Reflectance Sensor	QTRXL-HD-01A	Pololu	https://eckstein.com	1	3,60 €	\$4,14	3,60 €	\$4,14
14	Color Sensor	APDS 9960	Adafruit	https://www.adafruit.com	1	15,99 €	\$18,39	15,99 €	\$18,39
15	RGB-LED-Pixel	WS2812B	Worldsemi	https://leds-and.com	12	0,39 €	\$0,45	4,68 €	\$5,38
16	Motordriver	Motoron M2T256	Pololu	https://eckstein.com	2	34,18 €	\$39,31	68,36 €	\$78,61
17	Camera	8MP IMX219 175 Degree Ultra Wide Angle Raspberry Pi Camera Module	ArduCam	https://www.arducam.com	2	13,50 €	\$15,53	27,00 €	\$31,05
18	Switch	Kippschalter (blau) Rockerswitch	Funduino	https://funduino.com	6	1,26 €	\$1,45	7,56 €	\$8,69
19	LiPo	Akku-Pack, Li-Polymer, 11,1 V, 1500mAh	RED POWER	https://www.redpower.com	2	19,40 €	\$22,31	38,80 €	\$44,62
20	Servo	Savöx SH-0263MG+ SH-0263MG+ Mini	Savöx	https://www.savox.com	1	24,95 €	\$28,69	24,95 €	\$28,69
21							\$0,00	0,00 €	\$0,00
22	Chasis	Custom made, 3D-Printed Chasis	Felix Hafner, David Schumacher				\$0,00	0,00 €	\$0,00
23	Wheels	Custom made	Felix Hafner				\$0,00	0,00 €	\$0,00
24							\$0,00	0,00 €	\$0,00
25							\$0,00	0,00 €	\$0,00
26							\$0,00	0,00 €	\$0,00
27							\$0,00	0,00 €	\$0,00
28	Components, that were already available in our educational institution						\$0,00	0,00 €	\$0,00
29	AO3400A N-MOSFET						\$0,00	0,00 €	\$0,00
30	Resistors						\$0,00	0,00 €	\$0,00
31	Capasitors						\$0,00	0,00 €	\$0,00
32	Cables						\$0,00	0,00 €	\$0,00
33	Screws						\$0,00	0,00 €	\$0,00
34	Crimp material						\$0,00	0,00 €	\$0,00

(Price when purchased)

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#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local Currency)	Cost (USA Dollars)
1	Arduino & Embedded Development						
2	Arduino IDE	Software's Tool	Development environment for Arduino and Teensy programming	https://www.arduino.cc/en/software	Arduino	FREE	FREE
3	Neovim	Software's Tool	Highly configurable text editor for code development	https://neovim.io	Neovim Community	FREE	FREE
4	Teensy Board Manager	Software's Tool	Adds Teensy board support to Arduino IDE	https://www.pjrc.com/teensy/teensyduino.html	PJRC	FREE	FREE
5							
6	Embedded Libraries						
7	Wire	Library	I ² C communication library for microcontrollers	https://www.arduino.cc/reference/en/language/functions/communication/wire/	Arduino	FREE	FREE
8	SPI	Library	SPI communication interface library	https://www.arduino.cc/en/reference/SPI	Arduino	FREE	FREE
9	VL53L1X	Library	Time-of-Flight distance sensor driver library	https://github.com/pololu/vl53l1x-arduino	Pololu	FREE	FREE
10	VL53L8CX	Library	Multi-zone Time-of-Flight distance sensor library	https://github.com/STMicroelectronics/VL53L8CX_Sensor	STMicroelectronics	FREE	FREE
11	Adafruit APDS9960	Library	Proximity, gesture and ambient light sensor library	https://github.com/adafruit/Adafruit_APDS9960	Adafruit	FREE	FREE
12	Adafruit NeoPixel	Library	Control library for addressable RGB LEDs	https://github.com/adafruit/Adafruit_NeoPixel	Adafruit	FREE	FREE
13	Motoron	Library	Motor controller communication library	https://github.com/pololu/motoron-arduino	Pololu	FREE	FREE
14							
15	CAD & Hardware Design						
16	Fusion 360	Software's Tool	3D CAD modelling and mechanical design software	https://www.autodesk.com/products/fusion-360/overview	Autodesk	FREE (Educational)	FREE (Educational)
17	KiCad	Software's Tool	PCB design and electronics schematic software	https://www.kicad.org	KiCad Developers	FREE	FREE
18						FREE	FREE
19	Software Development Tools					FREE	FREE
20	Git	Software's Tool	Version control system for software development	https://git-scm.com	Software Freedom Conservancy	FREE	FREE
21	Python	Software's Tool	General-purpose programming language	https://www.python.org	Python Software Foundation	FREE	FREE
22							
23	Python Third-Party Libraries						
24	OpenCV (cv2)	Library	Computer vision and image processing library	https://opencv.org	OpenCV Team	FREE	FREE
25	NumPy	Library	Numerical computing and array processing library	https://numpy.org	NumPy Developers	FREE	FREE
26	PySerial	Library	Serial communication library for Python	https://pyserial.readthedocs.io	PySerial Developers	FREE	FREE
27	Basicmicro	Library	Motor controller communication library for robotics	https://www.basicmicro.com	Basicmicro	FREE	FREE
28							
29	Python Standard Libraries	Standard Libraries	Standard Python Libraries	https://docs.python.org/3/library	Python Software Foundation	FREE	FREE